The Value of Delayed Information: Predictive-Information Envelopes for Age-Limited Control

Envelopes, exact laws, and witnesses for age-limited decisions

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Working draft — v0.4 (battery v4 results integrated: dead zone observed, VAD reversal, one certified witness)

Abstract

A controller that acts on information of age D — a sensing delay, a telemetry lag, a stale cache — can outperform the best static policy only to the extent that the delayed observation still *predicts* the current state. Our companion systems work [1] proves this exactly for symmetric two-state Markov capacity ($\Delta(D) = \frac{1}{2}r(D)$, the autocorrelation at the sensing lag) with a β-mixing converse. This paper supplies the general theory. We (i) isolate the exact object — the *Bayes envelope* $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$, the expected excess value of posterior over prior beliefs; (ii) prove the universal *predictive-information envelope* $0 \leq \Delta_Y \leq \Omega\sqrt{\frac{1}{2}I(X;Y)}$ for any bounded objective with oscillation Ω, by Pinsker's inequality applied to the posterior process; (iii) derive the *bit-budget corollary*: a telemetry path delivering b bits per decision caps feedback value at $\Omega\sqrt{(b\ln 2)/2}$ regardless of what is encoded — compression can preserve but never create feedback value; (iv) show the envelope is *tight to leading order* in the canonical case: for the symmetric binary chain $I = \frac{r^2}{2} + O(r^4)$, so the bound reads $\Omega(\frac{|r|}{2} + O(|r|^3))$ against the exact $\frac{1}{2}|r|$; (v) prove by a two-bit construction that no universal equality $\Delta = f(I)$ can hold: mutual information is a scalar, decision value is a geometry, and two observations with identical I can carry all or none of the decision-relevant distinction; and (vi) give an exactness theory for binary state in which the local decision geometry at the prior sets the correlation exponent: for finite action sets the value of a delayed sample is exactly a sum of *hinges* in the lag covariance with sign-dependent dead zones; for smooth objectives it is exactly quadratic (squared error: $\Delta = C^2/\pi(1-\pi)$); the companion's linear law is the prior-on-kink boundary case. A nine-series battery (acoustics, bioacoustics, space weather, seismology, plus synthetic controls) finds no detectable history excess (equivalence-bounded within a registered margin) on the series where lag-1-matched surrogates predict none, observes the envelope's predicted $O(r^2)$ tightness on real data, and inverts the theorem into confidence-certified lower bounds on history information where the processes are non-Markov.

1 Introduction

Everywhere a controller acts on stale state — a scheduler admitting jobs against capacity sensed seconds ago, a trader on delayed quotes, a relay on an outdated channel estimate — the same question recurs: *what is the value of information of age D?* The age-of-information literature

[7] measures freshness; freshness, however, has no decision semantics: a maximally fresh sample of an i.i.d. process is worthless, and a stale sample of a slowly-mixing one is nearly as good as the present. Our companion systems paper [1] made this precise in one setting, proving that for symmetric two-state Markov capacity the advantage of age-D feedback over the best static policy is *exactly* $\Delta(D) = \frac{1}{2}r(D)$ — the autocorrelation at the sensing lag — and measuring the law on a live GPU cluster. This paper supplies the theory that result belongs to. The thesis is one line:

Age matters only through predictive information, and predictive information matters only through the Bayes envelope.

Concretely: (**C1**) we isolate the exact object — the Bayes envelope $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$, classical in decision theory [2, 4, 3] — and prove the universal *predictive-information envelope* $\Delta_Y \leq \Omega\sqrt{I(X;Y)/2}$ (Theorem 2), with the *bit-budget corollary*: b bits of telemetry per decision cap feedback value at $\Omega\sqrt{(b \ln 2)/2}$, so compression can preserve but never create feedback value. (**C2**) We show the envelope is *sharp where it matters*: on the canonical binary chain it matches the exact law to $O(r^3)$ (Proposition 5) — and the battery of §7 *observes* this tightness on real data, including the predicted $O(r^2)$ correction. (**C3**) We prove no universal equality $\Delta = f(I)$ exists for bounded objectives (Proposition 6): information is a scalar, value is a geometry. This sharpens against the classical exception — for *logarithmic* (unbounded) utility, Kelly and Barron–Cover [5, 6] show the value of side information *equals* I; boundedness is exactly what breaks equality down to an envelope. (**C4**) We give an exactness theory for binary state whose organizing fact is that *the local geometry of the value function at the prior sets the correlation exponent.* For finite action sets (polyhedral value functions) the value of a delayed sample is exactly a sum of *hinges* in the lag-D covariance, with sign-dependent dead-zone thresholds — linear response beyond, identically zero within; for smooth objectives (e.g. squared-error prediction, where exactly $\Delta = C(D)^2/\pi(1-\pi)$) the response is *quadratic*; and a prior sitting on a kink — the companion's symmetric case — is the unique configuration with threshold-free linear response (§6). The dead zone is the sharp polyhedral form of the Radner–Stiglitz nonconcavity of the value of information [13]: for asymmetric states under a polyhedral objective, weakly-correlated feedback is *worthless*, not merely weak, and the sign of the correlation matters. (**C5**) We order the three envelopes (mixing, information, bit-budget) below the master total-variation envelope and prove mixing and information are incomparable — for every lag and every K, a *preview process* keeps $\beta(D) \geq \frac{1}{2}$ while the information envelope, and the true value, vanish with K (§5). (**C6**) A nine-series battery across four physical domains bounds any undetected history excess within a registered equivalence margin where surrogates predict none, observes the envelope's tightness, and — read backwards — uses the theorem as a certified witness of history information on non-Markov series, always through a one-sided lower confidence bound: $I(X_t; \text{history}) \geq 2[(\widehat{\Delta} - \varepsilon)^+]^2$ (§7).

The scope is stated precisely: this is a unified theory for *exogenous one-step bounded-reward decisions* under delay — the action does not move the state, the reward is single-stage. Within that scope the theory is closed on itself: one decision-theoretic object explaining an exact systems law, envelopes ordered by what they require (a model, samples, or a bit rate), sharpness results tying envelope to law, impossibility results delimiting both, and a falsifiable empirical program run on real processes from four physical domains.

2 Setting and the exact object

Let X be the current exogenous state taking values in a finite set $\mathcal{X}$, with marginal $\bar{\mu}$. A controller observes a delayed quantity Y (a delayed sample X_{t-D}, a delayed history, or any $\sigma(\mathcal{F}_{\leq t-D})$-measurable statistic), chooses an action w, and receives a bounded one-step reward $R(w, x)$ with oscillation $\Omega = \max_{w,x} R - \min_{w,x} R$. Define the value of a belief $\mu \in \Delta(\mathcal{X})$:

$$\psi(\mu) \;=\; \max_{w} \sum_{x} R(w,x)\, \mu(x),$$

a pointwise maximum of linear functionals, hence convex; and the posterior $\mu_Y = \mathbb{P}(X \in \cdot \mid Y)$, with $\mathbb{E}[\mu_Y] = \bar{\mu}$ by the tower property. The optimal Y-measurable controller attains $V_Y^\star = \mathbb{E}[\psi(\mu_Y)]$; the best static controller attains $V_{\text{stat}} = \psi(\bar{\mu})$. The exact object of this paper is the **Bayes envelope**

$$\Delta_Y \;=\; V_Y^\star - V_{\text{stat}} \;=\; \mathbb{E}\big[\psi(\mu_Y)\big] - \psi(\bar{\mu}) \;\geq\; 0, \tag{1}$$

nonnegative by Jensen. Everything that follows is an envelope on (1); the companion paper's exact law is the case where (1) admits a closed form.

Lemma 1 (Lipschitz value [1, App. A]). $|\psi(\mu) - \psi(\mu')| \leq \Omega\, \|\mu - \mu'\|_{\text{TV}}$, *hence* $\Delta_Y \leq \Omega\, \mathbb{E}\|\mu_Y - \bar{\mu}\|_{\text{TV}}$.

The total-variation route specializes to the β-mixing converse of [1] when Y is the full delayed history: $\Delta(D) \leq \Omega\, \beta(D)$.

3 The predictive-information envelope

Theorem 2 (Predictive-information bound). *For any bounded objective with oscillation Ω and any delayed observable Y,*

$$0 \;\leq\; \Delta_Y \;\leq\; \Omega \sqrt{\tfrac{1}{2}\, I(X;Y)} \qquad \textit{(I in nats; in bits, } \Omega\sqrt{\tfrac{\ln 2}{2}\, I_{\text{bits}}}\textit{)}. \tag{2}$$

Proof. By Lemma 1, $\Delta_Y \leq \Omega\, \mathbb{E}\|\mu_Y - \bar{\mu}\|_{\text{TV}}$. Pinsker's inequality gives, for every realization y, $\|\mu_y - \bar{\mu}\|_{\text{TV}} \leq \sqrt{\tfrac{1}{2} D_{\text{KL}}(\mu_y \| \bar{\mu})}$. Taking the expectation over Y and applying Jensen to the concave square root,

$$\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\text{TV}} \;\leq\; \mathbb{E}\sqrt{\tfrac{1}{2} D_{\text{KL}}(\mu_Y \| \bar{\mu})} \;\leq\; \sqrt{\tfrac{1}{2}\, \mathbb{E}\, D_{\text{KL}}(\mu_Y \| \bar{\mu})} \;=\; \sqrt{\tfrac{1}{2}\, I(X;Y)},$$

the last equality being the definition of mutual information as the expected KL divergence of posterior from prior. □

Corollary 3 (Bit-limited telemetry). *Suppose the delivered observation satisfies $H(Y) \leq b \ln 2$ nats (in particular, whenever Y takes at most 2^b values, or is the output of a prefix code of expected length $\leq b$ bits). Then*

$$\Delta_Y \;\leq\; \Omega\, \min\Big\{1,\; \sqrt{(b \ln 2)/2}\Big\},$$

the clip because the trivial range bound $\Delta_Y \leq \Omega$ beats the square root already at $b \geq 3$ bits.

Lemma 4 (Garbling never helps: value monotonicity). *If $Y = g(Z)$ (deterministically or through any channel acting on Z alone), then $\Delta_Y \leq \Delta_Z$, and also $\Delta_Y \leq \Omega\sqrt{\tfrac{1}{2} I(X;Z)}$.*

Proof. $\mu_Y = \mathbb{E}[\mu_Z \mid Y]$ by the tower property, so convexity of ψ and Jensen give $\mathbb{E}\psi(\mu_Y) \leq \mathbb{E}\psi(\mu_Z)$; subtract $\psi(\bar{\mu})$. The second claim is Theorem 2 plus data processing ($I(X;Y) \leq I(X;Z)$). Note the two statements are different in kind: data processing alone bounds only the information *envelope*; the Jensen step bounds the *value itself* — the Blackwell comparison-of- experiments order [12] specialized to garbled delayed sensing. Together: compression can preserve feedback value; no encoding of the delayed signal can create it. □

Proposition 5 (Leading-order tightness in the canonical case)**.** *For a stationary fair binary pair with lag-D correlation $r = r(D)$,*

$$I(X_t; X_{t-D}) = \tfrac{1}{2}\big[(1+r)\ln(1+r) + (1-r)\ln(1-r)\big] = \tfrac{r^2}{2} + \tfrac{r^4}{12} + O(r^6),$$

an even function of r, and the exact single-sample value of the $0/1$ prediction objective is $\Delta(D) = \frac{1}{2}|r(D)|$ (negative correlation is exploited by reversing the prediction). The right side of (2) equals $\Omega\big(\frac{|r|}{2} + O(|r|^3)\big)$: the envelope matches the exact law to leading order at $\Omega = 1$. (The companion's persistent-chain assumption — flip probability $\le \frac{1}{2}$ — gives $r \ge 0$ and its statement $\frac{1}{2}r$; we carry the absolute value throughout.) The information envelope is not loose where it matters.

Proof. The joint distribution places mass $\frac{1+r}{4}$ on each agreeing pair and $\frac{1-r}{4}$ on each disagreeing pair; the stated I follows by direct computation, and the expansion from $(1 \pm r)\ln(1 \pm r) = \pm r + \frac{r^2}{2} \mp \frac{r^3}{6} + \cdots$. Then $\sqrt{\frac{1}{2}I} = \frac{r}{2}\sqrt{1 + \frac{r^2}{6} + O(r^4)}$. □

4 No universal equality $\Delta = f(I)$

Proposition 6 (Information is a scalar; value is a geometry)**.** *There is a state space, a bounded objective, and two observations Y_1, Y_2 with $I(X; Y_1) = I(X; Y_2) > 0$ but $\Delta_{Y_1} = \frac{\Omega}{2} > 0 = \Delta_{Y_2}$. Hence no function f with $\Delta_Y = f(I(X;Y))$ exists for all experiments.*

Proof. Let $X = (B_1, B_2)$ be two independent fair bits and let the objective depend only on B_1: actions $w \in \{0,1\}$, $R(w,x) = \Omega\,\mathbf{1}[w = B_1]$ (so the oscillation is Ω). Static value: $\Omega/2$. Let $Y_1 = B_1$ and $Y_2 = B_2$. Both have $I(X; Y_i) = \ln 2$. Observing Y_1 yields $V^\star = \Omega$, so $\Delta_{Y_1} = \Omega/2$; observing Y_2 leaves the posterior over B_1 uniform, so $\Delta_{Y_2} = 0$. The two experiments share every information quantity in $I(X;Y)$ yet differ maximally in value: the envelope (2) is attained within a constant on one and vacuous on the other. □

Remark 7. *This is the precise sense in which the companion paper's binary law is special: there the state is one decision-relevant bit, so the geometry collapses to a one-parameter family and the envelope inverts to an exact closed form. The general exact object remains (1).*

5 Envelope comparison

The three envelopes are not interchangeable, and their relationship is an ordered pair of relaxations of one master quantity.

Proposition 8 (Master envelope and universal ordering)**.** *For any delayed observable Y measurable w.r.t. the delayed history $\mathcal{F}_{\le t-D}$,*

$$\Delta_Y \;\le\; \Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}} \;\le\; \Omega\,\min\Big\{\beta(D),\; \sqrt{\tfrac{1}{2}I(X;Y)}\Big\},$$

where $\beta(D)$ is the β-mixing coefficient of the state process at lag D. The expected posterior movement $\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}}$ is the master envelope; the mixing and information envelopes are its two relaxations.

Proof. The first inequality is Lemma 1. For the second: since Y is $\mathcal{F}_{\le t-D}$-measurable and $\sigma(X) \subseteq \mathcal{F}_{\ge t}$, conditioning on the coarser Y can only shrink expected posterior movement, and $\mathbb{E}\|\mu_{\mathcal{F}_{\le t-D}} - \bar{\mu}\|_{\mathrm{TV}} \le \beta(D)$ because the β-coefficient takes its supremum over the larger future field [1, App. A]. The information relaxation is the Pinsker step in the proof of Theorem 2. □

Throughout, we use the companion's event-supremum convention

$$\beta(D) \;=\; \mathbb{E}\Big[\, \sup_{B\in\mathcal{F}_{\geq t}} \big|\mathbb{P}(B \mid \mathcal{F}_{\leq t-D}) - \mathbb{P}(B)\big| \,\Big],$$

which lower-bounds the absolute-regularity coefficient; conventions differ by factors of two across the literature, and every displayed constant below is tied to this one. Lower bounds on this β are what vacuousness of the mixing envelope requires.

Proposition 9 (The two relaxations are incomparable)**.** *Neither relaxation dominates the other.*

(i) Mixing strictly tighter: *for the symmetric two-state chain with $Y = X_{t-D}$, $\beta(D) = \frac{1}{2}|r|$ while $\sqrt{\frac{1}{2}I} = \frac{|r|}{2}\sqrt{1 + \frac{r^2}{6} + O(r^4)} > \beta(D)$ for every $|r| \in (0,1)$.*

(ii) Information arbitrarily tighter: *for every fixed lag D and every $K \geq 2$ there is a stationary state process and bounded objective with*

$$\beta(D) \;\geq\; \tfrac{1}{2}, \qquad I(X_t; \mathcal{F}_{\leq t-D}) \;\leq\; h_2(1/K) + \tfrac{\ln 2}{K} \;\xrightarrow[K\to\infty]{}\; 0, \qquad \Delta \;=\; \tfrac{\Omega}{2K},$$

where h_2 is the binary entropy in nats: the mixing envelope stays $\geq \Omega/2$ while the information envelope, and the true value, vanish.

Proof. (i) is Proposition 5 together with $\beta(D) = \frac{1}{2}|r(D)|$ for the binary chain (Appendix A). (ii) *The preview process (one per (D,K)).* Fix D and K, set $m = D + K$. Let c_t be i.i.d. fair bits, let $\Theta \sim \mathrm{Unif}\{0,\ldots,K-1\}$ be an independent phase (randomized to make the pair stationary), and let the observable state be $X_t = (c_t, p_t)$ with

$$p_t \;=\; \begin{cases} c_{t+m} & t \equiv \Theta \pmod{K} \\ \perp & \text{otherwise,} \end{cases}$$

a sensor that occasionally publishes a *forecast* m steps ahead. The objective depends only on the current payoff bit: $R(w,x) = \Omega\,\mathbf{1}[w = c]$. *Mixing (lower bound).* The window $[t-m,\, t-D]$ has length K, hence contains exactly one preview slot $s \equiv \Theta$; that preview is in the delayed past and reveals c_{s+m} with $s+m \in [t,\, t+K]$, a present-or-future payoff bit. The event $B = \{c_{s+m} = 1\} \in \mathcal{F}_{\geq t}$ then has conditional probability in $\{0,1\}$ against marginal $\frac{1}{2}$, so $\beta(D) \geq \frac{1}{2}$. (Previews shared between past and future create further dependence; we claim only the lower bound, which is what vacuousness of the mixing envelope needs.) *Information (upper bound).* X_t is determined by the triple (payoff bit c_t; slot indicator $\mathbf{1}[t \equiv \Theta]$; preview value when present, a *future* bit independent of the past). The delayed history reveals the slot indicator (it determines Θ) and reveals c_t itself exactly when $t - m \equiv \Theta$ (probability $1/K$); hence $I(X_t; \mathcal{F}_{\leq t-D}) \leq h_2(1/K) + \frac{\ln 2}{K}$. *Value.* Feedback wins $\Omega/2$ on the revealed fraction $1/K$ and nothing elsewhere: $\Delta = \Omega/(2K)$. The information envelope $\Omega\sqrt{(h_2(1/K) + \ln 2/K)/2}$ decays with K (rate $\widetilde{O}(K^{-1/2})$ against the true K^{-1} — loose, but not blind); the mixing envelope does not decay at all. □

Remark 10 (Why keep the information form at all)**.** *Proposition 8 shows the TV master envelope is always at least as tight, and (i) shows mixing can beat information. Three properties are the information envelope's own: it is* estimable *— the plug-in $\hat{I}(X_t; X_{t-D})$ in §7 needs only pair counts, while the β-coefficient's supremum over the future σ-field is not estimable without a process model; it* composes with channels *— data processing gives Corollary 3, and no analogous statement exists for β (mixing has no channel semantics); and it* adds across sensors *via the chain rule, giving budget statements for multi-sensor telemetry. In short: mixing is the sharper analysis tool when a model is in hand; information is the operational one when only data or a bit budget is.*

Envelope	Bound on Δ_Y	Needs	Blind spot
master (TV)	$\Omega\,\mathbb{E}\|\mu_Y - \bar{\mu}\|_{\mathrm{TV}}$	posterior process	not estimable directly
mixing	$\Omega\,\beta(D)$	process model	future events beyond X_t (Prop. 9ii)
information	$\Omega\sqrt{I(X;Y)/2}$	pair samples	decision geometry (Prop. 6)
bit budget	$\Omega\sqrt{(b\ln 2)/2}$	channel rate only	everything but the rate

Table 1: The envelope family. The structure is a *branching*, not a hierarchy: both mixing and information relax the master TV envelope, in incomparable directions (Proposition 9), and the bit budget further relaxes the information branch via Lemma 4: TV $\to$ { mixing; information $\to$ bit budget }. Each branch trades knowledge for tightness in its own currency; the battery of §7 measures the first three on real processes.

6 Exactness: a complete account for binary state, and its limits

When does the Bayes envelope admit a closed form — an exact *law* rather than an envelope? For binary state the answer is: always, and the form is a hinge.

Theorem 11 (The hinge law, polyhedral case)**.** *Let $X_t \in \{0,1\}$ be stationary with $\pi = \mathbb{P}(X_t{=}1) \in (0,1)$, let $Y = X_{t-D}$, and write $C = C(D) = \mathrm{Cov}(X_t, X_{t-D})$. Assume the action set is finite — or, more generally, that the value function $\psi(p) = \sup_w[pR(w,1) + (1-p)R(w,0)]$ is* polyhedral *(a maximum of finitely many affine functions of the belief), with kinks $p_1^* < \cdots < p_K^*$ and slope jumps $\Delta s_k > 0$. Then, exactly,*

$$\Delta(C) \;=\; \begin{cases} \displaystyle\sum_k \Delta s_k\,\big(C - a_k^+\big)^+, & C \ge 0, \\ \displaystyle\sum_k \Delta s_k\,\big((-C) - a_k^-\big)^+, & C < 0, \end{cases} \qquad a_k^+ = \begin{cases} (p_k^* - \pi)\,\pi, & p_k^* \ge \pi, \\ (\pi - p_k^*)(1-\pi), & p_k^* < \pi, \end{cases} \quad a_k^- = \begin{cases} (p_k^* - \pi)(1-\pi), & p_k^* \ge \pi, \\ (\pi - p_k^*)\,\pi, & p_k^* < \pi. \end{cases}$$

Δ is piecewise-linear and nondecreasing in $|C|$ on each covariance-sign branch*, identically zero on the branch dead zone; unless $\pi = \frac{1}{2}$ or the kink set is symmetric about π, the two branches differ — positive and negative correlation of the same magnitude can have different value, one of them zero.*

Proof. Write $\psi(p) = \psi(p_1^*) + s_0(p - p_1^*) + \sum_k \Delta s_k (p - p_k^*)^+$ (polyhedral by hypothesis). Observing Y produces the two-point posterior $p_1 = \pi + C/\pi$ (w.p. π) and $p_0 = \pi - C/(1-\pi)$ (w.p. $1-\pi$), whose mean is π; the affine part of ψ cancels in $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi)$, leaving $\Delta = \sum_k \Delta s_k \{\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*)^+\}$. Fix k. For $C \ge 0$ ($p_0 \le \pi \le p_1$): if $p_k^* \ge \pi$, $\mathbb{E}(p_Y - p_k^*)^+ = \pi(p_1 - p_k^*)^+ = (C - (p_k^* - \pi)\pi)^+$ and $(\pi - p_k^*)^+ = 0$; if $p_k^* < \pi$, the pointwise identity $(x)^+ - x = (-x)^+$ gives $\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*) = \mathbb{E}(p_k^* - p_Y)^+ = (1-\pi)(p_k^* - p_0)^+ = (C - (\pi - p_k^*)(1-\pi))^+$. For $C < 0$ the posteriors trade places ($p_1 < \pi < p_0$): if $p_k^* \ge \pi$, only p_0 can exceed the kink and $\mathbb{E}(p_Y - p_k^*)^+ = (1-\pi)(p_0 - p_k^*)^+ = ((-C) - (p_k^* - \pi)(1-\pi))^+$; if $p_k^* < \pi$, the same identity gives $\pi(p_k^* - p_1)^+ = ((-C) - (\pi - p_k^*)\pi)^+$. Summing over k gives the two branches. $\square$

6.1 Smooth objectives: quadratic response, and the trichotomy

The polyhedral hypothesis is not decoration. For a continuous action set ψ is a supremum of *infinitely* many affine functions and need not be piecewise-linear, and the hinge law fails — constructively:

Proposition 12 (Squared-error prediction: exactly quadratic)**.** *For $w \in [0,1]$ and $R(w,x) = -(w-x)^2$ (oscillation $\Omega = 1$), $\psi(p) = -p(1-p)$ is smooth and strictly convex, and, exactly,*

$$\Delta(D) \;=\; \mathrm{Var}(p_Y) \;=\; \frac{C(D)^2}{\pi(1-\pi)}:$$

quadratic in the covariance, no hinge, no dead zone, for every π. More generally, whenever ψ is twice continuously differentiable near π,

$$\Delta(D) \;=\; \tfrac{1}{2}\,\psi''(\pi)\,\mathrm{Var}(p_Y) + o(C^2) \;=\; \frac{\psi''(\pi)}{2\pi(1-\pi)}\,C(D)^2 + o\big(C(D)^2\big).$$

Proof. The optimal action under belief p is $w = p$, so $\psi(p) = -p(1-p) = p^2 - p$; then $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi) = \mathbb{E}[p_Y^2] - \pi^2 = \mathrm{Var}(p_Y)$, exact because ψ is quadratic, and $\mathrm{Var}(p_Y) = \pi(C/\pi)^2 + (1-\pi)\big(C/(1-\pi)\big)^2 = C^2/\pi(1-\pi)$. The general statement is the second-order Taylor expansion of ψ at π against the mean-π posterior spread. □

Remark 13 (The trichotomy: local decision geometry sets the correlation exponent)**.** *The cases exhaust the local behaviours of a convex ψ at the prior, and each fixes how weak dependence buys value:* prior on a kink $\Rightarrow$ *linear response $\Delta \propto |C|$;* prior interior to a flat piece $\Rightarrow$ *dead zone, $\Delta = 0$ until a threshold;* smooth curvature $\Rightarrow$ *quadratic response $\Delta \propto C^2$. The companion's law is the first case — symmetry puts the prior on the decision kink, the unique threshold-free polyhedral configuration. The dead zone is the sharp polyhedral form of the Radner–Stiglitz nonconcavity of the value of information [13], whose smooth setting is our third case (zero marginal value at zero information, but not identically zero). And the sign-dependence of Theorem 11 means even the* direction *of correlation matters away from symmetry: with $\pi = 0.2$ and one kink at $p^* = 0.3$, covariance $+0.03$ clears its threshold $(0.1)(0.2) = 0.02$ and has value, while -0.03 sits inside its threshold $(0.1)(0.8) = 0.08$ and is worthless.*

Corollary 14 (The companion law, and why it is threshold-free)**.** *For the symmetric chain ($\pi = \frac{1}{2}$) with the $0/1$ prediction objective ($K{=}1$, $p^* = \frac{1}{2}$, $\Delta s = 2$): $C^* = 0$ and $C(D) = \frac{1}{4}r(D)$, so $\Delta(D) = 2 \cdot \frac{1}{4}r(D) = \frac{1}{2}r(D)$ — the exact law of [1]. Symmetry places the prior* on *the kink, which is the unique configuration with no dead zone; the linear-in-r law is the degenerate hinge.*

Remark 15 (The dead zone is a prediction, not a curiosity)**.** *For $\pi \neq p^*$ the hinge says weakly-correlated delayed feedback is* worthless *— not slightly valuable — until $|C(D)|$ crosses C^*: below threshold, both posteriors fall on one side of the decision kink and the optimal action never changes. Operationally: a rare-event admission controller (π small, p^* moderate) gains* nothing *from telemetry until its sensing age satisfies $|C(D)| > (p^* - \pi)\pi$ — an exact, testable cutoff for when to stop paying for freshness at all. None of the envelopes of §5 see this: the dead zone is a fact about geometry (posteriors vs. kink), which is precisely what scalar information summaries discard (Proposition 6).*

Remark 16 (Beyond two states: the layer-cake lift, and the boundary)**.** *The companion's layer-cake result [1, Prop. 2] lifts binary exactness to capacities $a_t \in \{0,\dots,C\}$ with monotone threshold objectives: the objective decomposes over layer indicators $x_t^u = \mathbf{1}[a_t \geq u]$ and the total advantage is the exact sum $\Delta(D) = \sum_u \pi_u r_u(D)$ — each layer a binary experiment, each governed by its hinge (there with symmetric thresholds, so zero dead zones). The boundary of exactness is now visible from both sides: exact laws hold precisely when the posterior family is effectively one-dimensional (a scalar belief per layer, moved along a line by the observation); the two-bit construction of Proposition 6 is the minimal experiment whose posterior geometry is genuinely two-dimensional, and there the value ceases to be a function of any scalar summary of dependence — correlation or information alike.*

7 The law and its envelopes across domains

The companion paper had room for one paragraph of this study; the full protocol and results are reported here. The question has two halves. First, *single-sample sufficiency*: on real binary state processes, does history beyond the single delayed sample add decision value beyond the *exact empirical single-sample Bayes value* — computed from the pooled 2×2 lag table with no symmetry assumption, so that Theorem 11's imbalance effects (dead zones included) are carried automatically? Second, *the envelope*: does the measured value respect, and how tightly does it approach, the predictive-information bound of Theorem 2?

7.1 Protocol

States. Binary regime series from four physical domains plus controls: acoustic source azimuth and voice activity (LOCATA, 120 Hz); sperm-whale coda type (Dominica corpus, per-exchange); GOES space-weather series (X-ray flux, magnetometer, proton flux; 1–5-minute cadence); seismic station regime (ANMO, 1-minute); a synthetic two-state Markov chain (flip probability 0.02) as positive reference; an i.i.d. fair-bit series as null. Each series is binarized by the domain's registered thresholding (the companion study's preparation scripts, unchanged).

The two estimands. For each lag D on a per-domain grid: (i) the *exact empirical single-sample Bayes value* $\widehat{V}_1(D) = \sum_y \hat{w}_y \max(\hat{p}_y, 1-\hat{p}_y) - \max(\hat{\pi}, 1-\hat{\pi})$, the plug-in of (1) for $Y = X_{t-D}$ computed from the pooled lag table — for a balanced series this is $\frac{1}{2}|r(D)|$, and for an imbalanced one it carries Theorem 11's thresholds automatically (the per-domain class proportion $\hat{\pi}$ is reported alongside); (ii) the value of the best *delayed-history* policy: a k=6-lag lookup predictor trained and evaluated by blocked cross-validation (5 folds, contiguous blocks), reported as accuracy above the *best static (majority) policy* $\max(\hat{\pi}, 1-\hat{\pi})$ — the correct static benchmark; accuracy above $\frac{1}{2}$ overstates value whenever $\hat{\pi} \neq \frac{1}{2}$. The *history gain* is the difference (ii)−(i).

Surrogate calibration. The null for "history adds nothing beyond the law" is a population of 60 Markov surrogates per domain, each matched to the real series' lag-1 transition probabilities and evaluated by the identical pipeline, with $M = 299$ surrogates. Significance uses the exact Monte Carlo form $p = (1 + \#\{T_j \geq T_{\text{obs}}\})/(M+1)$ (smallest reportable value $1/300 \approx 0.0033$; entries at that floor are written $p < .004$, never .00), together with the companion study's practical floor of 0.03 on the effect. Because each surrogate is Markov, its *true* history gain is zero and its own exact $\widehat{V}_1$ is computable from its lag table, so the M surrogate gain estimates form a genuine sample of the pipeline's *estimation error under matched dependence*; its 95th percentile ε_{dom} calibrates every one-sided bound below. *Non-rejection is not exactness*: where the test does not fire, we report the one-sided upper confidence bound $U = \text{gain}_{\max} + \varepsilon_{\text{dom}}$ and claim "no detectable excess (equivalent within the margin)" only when $U < 0.03$.

The envelope measurement. For each domain and lag, the plug-in mutual information $\widehat{I}(X_t; X_{t-D})$ from pooled pair counts (2×2; plug-in bias $\leq 3/(2N)$ nats, negligible at the pooled sample sizes reported) and the envelope $\sqrt{\widehat{I}/2}$ of Theorem 2. Two registered uses. *Forward* (envelope check): $\widehat{\Delta}_{\text{hist}}(D) \leq \sqrt{\widehat{I}(D)/2} + \varepsilon_{\text{dom}}$ — an apparent violation smaller than the matched-dependence noise is estimator noise, not physics. *Backward* (certified witness): a policy value is converted into an information lower bound only through its one-sided *lower* confidence bound,

$$I(X_t; \text{window}) \;\geq\; 2\left[(\widehat{\Delta}_{\text{hist}} - \varepsilon_{\text{dom}})^+\right]^2,$$

never through the raw point estimate, whose positive noise would inflate the certificate.

7.2 Results

Table 7.2 reports the corrected battery (v4): per domain, the class proportion $\hat{\pi}$; the maximum history gain over the exact single-sample Bayes value $\widehat{V}_1$; the exact Monte Carlo p (M=299; floor 1/300, written $< .004$); the matched-dependence noise bound ε_{dom}; the equivalence bound $U = \text{gain}_{\max} + \varepsilon_{\text{dom}}$ (margin 0.03); the maximum excess of the history policy over the single-lag envelope, with certification only when it clears ε_{dom}; and the certified window-information lower bound $2[(\widehat{\Delta}_{\text{hist}} - \varepsilon_{\text{dom}})^+]^2$ where applicable.

Domain	$\hat{\pi}$	$\text{gain}_{\max}$	p	ε_{dom}	U	env. excess	verdict
Two-state Markov (ref)	.470	+.000	.78	.006	.007	−.007	equivalent
i.i.d. null (control)	.497	+.012	.023	.011	.023	+.009	below floor
LOCATA azimuth	.494	+.008	.063	.009	.016	−.026	equivalent
LOCATA VAD	.636	−.000	.82	.052	.052	−.027	no excess detected†
Sperm-whale codas	.483	+.046	1.0	.117	.164	+.027	no excess detected†
GOES protons	.489	+.029	.11	.033	.063	+.021	no excess detected†
GOES X-ray flux	.421	+.028	$< .004$	.008	.036	−.008	detected, below floor
GOES magnetometer	.456	+.033	$< .004$	.010	.043	−.004	history adds (at floor)
Seismic ANMO	.508	+.158	$< .004$	.014	.171	+**.148**	history adds (certified)

endtabular

Table 2: Battery v4 (corrected protocol; `battery_v4.json`). "Equivalent" = $U < 0.03$; †"no excess detected" = test does not fire but $U \geq 0.03$, so equivalence is *not* established (noise-limited) — a distinction the uncorrected protocol could not express. Bold: the only envelope excess that clears the matched-dependence noise bound. Pooled pair counts range from 2.0k (protons) to 21k (azimuth) per lag.

Four readings, in increasing order of consequence.

The corrections bite exactly where the review predicted. Under the chance baseline (v3), four series appeared to carry history value beyond the single sample (gains 0.100–0.367, all "$p = .00$"). Under the majority baseline and exact $\widehat{V}_1$: LOCATA VAD's gain *vanishes* ($+0.100 \to -0.000$, $p = .82$; $\hat{\pi} = .636$ — the entire effect was the majority-class head start), GOES X-ray shrinks $0.185 \to 0.028$ (below the practical floor; $\hat{\pi} = .421$), magnetometer $0.147 \to 0.033$ (at the floor), and only seismic survives at strength ($0.367 \to 0.158$). Imbalance, not history, carried most of the apparent excess — and the sub-unity envelope ratios of v3 disappear with it (all balanced-domain tightness ratios now sit at 1.11–1.35, on the Pinsker side of unity, exactly as population theory requires).

Single-sample sufficiency is the rule. On eight of nine series the delayed single sample, evaluated exactly (majority baseline, dead zones included), captures all detectable feedback value at the battery's resolution — with the equivalence bound $U < 0.03$ certified for the balanced, well-sampled series (Markov reference, azimuth) and the weaker "no excess detected" verdict, honestly labelled, where noise is wide (short whale exchanges, sparse protons).

The dead zone is observed. The imbalanced GOES series realize Theorem 11 in the wild: X-ray ($\hat{\pi} = .421$, kink at $\frac{1}{2}$, threshold $C^* \approx 0.033$) has $\widehat{V}_1$ *exactly zero* across its moderate-covariance lags — the majority policy is unbeatable by any single delayed sample — while its plug-in information remains positive; its envelope-to-value ratio diverges ($\gg 10^6$) not as an artifact but as the theorem's geometry: *information without value*, Proposition 6 measured on a physical process. Freshness metrics, and information metrics alike, cannot see this; the Bayes envelope does.

The certified witness stands — once. Seismic regime is the one series where the history policy's excess over the single-lag envelope clears the matched-dependence noise bound (+0.148 vs.

$\varepsilon = .014$), and its confidence-certified window-information bound is $I(X_t; \text{window}) \geq 0.176$ nats — aftershock clustering carrying an order of magnitude more predictive information than the last sample alone ($\widehat{I}_1 \approx 0.002$ at the certifying lag). Magnetometer shows a small certified gain at the practical floor. The instrument reads in both directions, and under the corrected statistics its two directions are: equivalence, most places; certified history information, exactly where the physics says it should be.

8 Related work

Value of information. The Bayes envelope is the classical object of statistical decision theory [2, 4], and its convex geometry is charted territory: De Lara and Gossner [3] study the value function as a support function, characterize when information has positive value, and give local estimates near uninformative experiments. Our Lipschitz lemma instantiates their payoffs–beliefs duality; the hinge and dead-zone analysis of §6 is the exact delayed-binary specialization of that convex-analytic programme; the comparison-of-experiments order behind Lemma 4 is Blackwell's [12]; and the Radner–Stiglitz nonconcavity [13] anticipates the dead zone in smooth form. The information envelope itself is assembled from textbook inequalities — kindred moves appear wherever bounded value must be controlled by divergence, e.g. the information-ratio regret analyses of Russo and Van Roy [11]. What we have not found elsewhere is the assembly this paper is about: the envelope *anchored to an exact law* (sharp to $O(r^3)$, with the $O(r^2)$ correction observed on real data), its ordering against mixing envelopes with both separations constructed, and its inversion into a certified information witness.

The logarithmic exception. For log-utility gambling the value of side information *equals* mutual information: Kelly [5] for the growth rate, Barron and Cover [6] for the general bound $\Delta \leq I$ with equality characterizations. Log utility is unbounded, escaping our Ω-bounded class; the pair of results brackets the landscape — unbounded log utility buys equality, bounded utility only an envelope, and Proposition 6 shows the gap is essential, not technical.

Age of information, and its critics. AoI [7] treats freshness as the object, with sampling-policy results such as [8] optimizing age processes. That plain age is not decision significance is already argued *within* the field: age-of-incorrect-information replaces the clock with an error-aware penalty [14], explicit AoI-vs-VoI scheduling comparisons exist for networked control [15], and value-of-information metrics for feedback control have been quantified directly [16]. Our claim is accordingly not that age lacks decision semantics — that point is established — but sharper: for exogenous one-step bounded-reward decisions the exact semantic object is the posterior Bayes envelope, which admits a universal mutual-information converse, an explicit binary geometry (hinges, dead zones, quadratic regimes), and an operational inversion into predictive-information witnesses.

Delayed information in control. Delayed-sharing information structures go back to Witsenhausen [9] and are systematized by the common-information approach [10]; remote estimation studies the estimation face of the same trade [8]. Our setting is the tractable exogenous corner of that world — the action does not affect the state — which is exactly what makes closed forms and universal envelopes available.

Predictive information. Bialek, Nemenman and Tishby [17] study $I(\text{past}; \text{future})$ as a measure of process complexity. Theorems 2 and 11 are, in that vocabulary, statements about what predictive information can and cannot buy a bounded-utility decision maker — with the battery's witness bound $I \geq 2\widehat{\Delta}^2$ operating as a decision-theoretic estimator of predictive information from policy value alone.

9 Conclusion

The companion paper measured one exact law on one system. This paper locates that law inside a unified theory of exogenous one-step bounded-reward decisions under delay: value of delayed information is the Bayes envelope; the envelope is bounded by predictive information (universally), by mixing (given a model), and by the telemetry bit rate (given nothing); it is exactly a hinge in covariance for binary state, exactly a layered sum for monotone multi-state objectives, and exactly nothing — inside the dead zone — for asymmetric weakly-coupled sensing. Beyond one effective dimension, no scalar summary of dependence determines value at all. Each claim is falsifiable, and the nine-series battery exercises all of them: no history excess is detectable where the equivalence bound requires none, the envelope's predicted tightness appears with its $O(r^2)$ correction, and its violations, where the theory requires them, certify history information in nats. The instrument reads in both directions; that, more than any single inequality, is what we hope survives.

A Companion results restated

For self-containedness we restate the companion results used above, with proofs where short; [1] is under submission and unavailable to a reviewer.

Mixing coefficient. Throughout, $\beta(D) = \mathbb{E}[\sup_{B\in\mathcal{F}_{\geq t}} |\mathbb{P}(B \mid \mathcal{F}_{\leq t-D}) - \mathbb{P}(B)|]$ (event-supremum convention; a lower bound for the absolute-regularity coefficient, so the vacuousness results transfer conservatively).

Binary chain. For the symmetric two-state chain with flip probability p and $r(D) = (1-2p)^D$: conditioning on the delayed state moves the law of X_t by total variation $\frac{1}{2}|r(D)|$ from either value of the conditioner, and the Markov property collapses the past to the delayed sample; hence $\beta(D) = \frac{1}{2}|r(D)|$. The exact value of the 0/1 prediction objective is $\Delta(D) = \frac{1}{2}|r(D)|$: predict the delayed state if $r > 0$, its complement if $r < 0$; either beats the static $\frac{1}{2}$ by half the correlation magnitude.

Layer-cake lift (companion Prop. 2; statement and sketch). For capacity $a_t \in \{0, \dots, C\}$, threshold indicators $x_t^u = \mathbf{1}[a_t \geq u]$ with $\pi_u = \mathbb{P}(x_t^u{=}1)$, and objectives that are positive combinations of per-layer threshold rewards (goodput minus over-admission surplus in the companion's setting), the objective decomposes additively over layers via $\min(w, a) = \sum_u \mathbf{1}[w \geq u]\, x_t^u$ and $(w-a)^+ = \sum_u \mathbf{1}[w \geq u](1 - x_t^u)$; the age-$D$ optimum applies the per-layer binary policy (the delayed indicators, whose nesting is inherited from the observation), and the total advantage over the best comonotone static budget at matched surplus is $\Delta(D) = \sum_u \pi_u r_u(D)$, $r_u(D) = \mathrm{corr}(x_t^u, x_{t-D}^u)$.

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